

Design Digital Circuits Logically

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Students today have digital electronics as a part of their curriculum right from their high-school days. Unfortunately, in most cases, neither are they given hands-on experience nor are they exposed to simulations where they actually get to know how the entire system works.

In this article we discuss a logic circuit simulator called Logically that allows academicians to teach logic gates and digital circuits easily and effectively. A trial version of the latest release of this software is bundled in this month's DVD for EFY Plus.

How Logically helps students

The software is designed to cater to the needs of students learning logic design. Educators too find Logically beneficial while teaching theory in their classes. Let us take a quick look at the features that make the tool appealing to the academic community.

Smart UI. Logically comes up with a smart yet simple, intuitive user interface (UI). The user can drag and drop components, copy/paste or zoom these. Students learn by making mistakes. And that is alright as long as the undo option is just a click away. Students can customise, encapsulate, drag and drop designs, eliminating the duplication of time and effort. They can import libraries built and customised as per their curriculum.

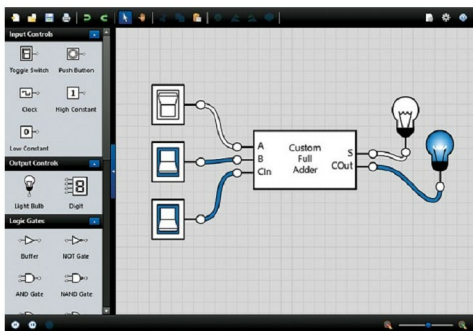
Control over debugging. The user can control the debugging process by pausing simulation. This helps in an effective learning by having a step-by-step analysis of the signal flow through each connection, control and module in the circuit.

Multi-platform. The software is compatible with multiple platforms. The choice on whether to go for Win-

Simulating a logic circuit, Full Adder

Let us study the case of a typical combinational logic circuit, a Full Adder. It adds two bits and a carry, and outputs a sum bit and a carry bit as shown in the figure. In order to simulate a Full Adder, the student has to drag three toggle switches, representing the two input bits and the input carry bit from Input Controls section. He can have a customised Full Adder circuit in the library, so that he can just drag and drop the module in the circuit. Two light bulbs representing the output carry bit and the sum bit can be borrowed from Output Controls section.

All these components can now be connected using connectors. Once connections are made, the circuit can be debugged to view the result. Outputs can be visualised for various high-low combinations of the input.



dows or Mac as the operating system to run this software is left to the user.

What is new in Logically

The introduction of integrated circuit (IC) libraries, tri-state buffers and buses and multiple-files-handling capability are among the latest enhancements that were made to this software.

Introduction of tri-state buffers.

This buffer acts as an ordinary buffer when enable input is high. In this case, the other input value is passed unchanged. Whereas, if enable input is low, output of the tri-state component will be a high-impedance (high-

Z) signal. This signal is received by a bus, which allows combining multiple connections into a single one.

If a high-Z signal is received by the bus component at one of its inputs, that input is ignored completely. Ideally, all inputs to the bus except one will have a high-Z value, and the input without high-Z value will be either high or low. This value will be passed on to the output. When there are both high and low inputs at the input of the bus at the same time, output will be an error indicated by an unknown signal. Addition of a buffer ensures timely updates of all inputs. If other